EE360 – Term 252 — Term Project Description

Medium Voltage Test Power System Design for a

4 kV, 1.551 MW Induction Motor

Submission Deadline: April 23rd, 2026

1. Project Overview

A maintenance workshop has received a **4 kV, 1.551 MW ABB induction motor** for overhaul. Before the motor is returned to its original facility, the workshop is required to:

- Energize the motor under controlled conditions
- Run it at no-load to verify mechanical integrity
- Confirm and document its operational performance

Figure 1: Motor Nameplate Data – ABB 4 kV, 1.551 MW Induction Motor

However, the workshop only has access to a **400 V diesel generator**. Despite having sufficient power capacity, this generator *cannot* directly supply the required medium-voltage level.

Engineering Challenge

Design a complete and practical system capable of operating a **4 kV induction motor from a 400 V diesel generator source**, based on real industrial components and current market constraints.

2. Project Objective

Design a complete **Medium Voltage (MV) Test Power System** that enables:

- Safe and controlled energization of the 4 kV motor
- Stable no-load testing under diesel generator supply
- Reliable and repeatable operational verification

The proposed solution must be:

- **Technically sound** – complying with relevant IEC standards
- **Commercially feasible** – based on real, market-available industrial equipment
- **Operationally practical** – implementable within a workshop environment

3. Scope of Work

3.1. System Design

Students must design a complete MV test power system, including the following principal components:

- **Step-up transformer** – 0.4 kV to 4 kV voltage transformation
- **MV switchgear** – protection, isolation, and control at the MV bus
- **Motor starting system** – to manage and limit the starting inrush current

3.2. Single Line Diagram (SLD)

Develop a complete and clearly labeled **Single Line Diagram (SLD)** that illustrates the entire power flow path, including:

- Diesel generator (LV source)
- Step-up transformer (LV/MV)
- MV switchgear and protection devices
- Motor starter or Variable Frequency Drive (VFD)
- 4 kV induction motor

3.3. Starting Methods Evaluation

Students must evaluate **at least two** of the following motor starting methods and compare their suitability for this application:

1. **Autotransformer Starter** – reduced-voltage starting via tap selection
2. **MV Soft Starter** – electronic voltage ramp-up with thyristor control
3. **MV Variable Frequency Drive (VFD)** – full variable-speed control with smooth acceleration and regenerative capability

4. Engineering Analysis

4.1. Motor Base Calculations

Students must establish all relevant electrical base quantities for the motor – including rated current, rated power factor, synchronous speed, and per-unit impedance values – using the nameplate data provided in Figure 1.

4.2. Required Analysis

The following engineering analyses are mandatory:

- **Starting current estimation** – calculation of full-voltage inrush current and reduced-voltage starting current for each method
- **Voltage drop analysis** – voltage deviation at the generator terminals and MV bus during motor start
- **Transformer sizing** – selection of appropriate kVA rating, impedance, and cooling class
- **Generator loading analysis** – verification of generator adequacy under starting and running conditions
- **Short-circuit level estimation** – fault current assessment at the MV bus to guide switchgear selection

4.3. Generator–Motor Interaction

Students must rigorously analyze the dynamic interaction between the generator and motor, covering:

- **Voltage dip during motor starting** – magnitude and duration of the voltage depression at the generator terminals
- **Transformer energization inrush** – magnetizing inrush current effect on generator voltage stability

- **Generator stability and limitations** – assessment of whether the generator can maintain stable voltage regulation throughout the starting transient

Important Note

Starting current may reach up to **6 times** the motor rated current under direct-on-line (DOL) starting conditions. This must be accounted for in all sizing and stability calculations.

5. Standards and Compliance

The design must align with the following international IEC standards, which may be referenced throughout the analysis and documentation:

- **IEC 60034** – Rotating Electrical Machines
- **IEC 60076** – Power Transformers
- **IEC 62271** – High-Voltage Switchgear and Controlgear

6. Simulation Requirement

Students must model and simulate the complete MV test system using:

- **MATLAB/Simulink** – dynamic system modeling and time-domain simulation
- **Python** – numerical modeling using `scipy` or `numpy` for steady-state and transient calculations

The simulation must demonstrate and compare the following:

- Motor starting behavior: speed, torque, and current vs. time
- Generator terminal voltage response during the motor start transient
- Side-by-side comparison of the selected starting methods

7. Commercial Analysis

7.1. Vendor Selection

Students must propose real, commercially available equipment from reputable industrial vendors, including but not limited to: **ABB**, **Siemens**, and **Schneider Electric**. All proposed equipment must be identifiable by vendor catalog reference or model number.

7.2. Cost Evaluation

Students must estimate the following cost elements for each starting method:

- **CAPEX (Capital Expenditure)** – equipment procurement costs
- **Installation complexity** – relative installation effort, civil works, and commissioning considerations

7.3. Trade-Off Analysis

A structured trade-off comparison must be presented for all evaluated starting methods across the following criteria:

Criterion	Description
Technical Feasibility	Ability to operate the motor safely and reliably
Generator Compatibility	Voltage stability and generator loading during start
Capital Cost	Equipment and procurement expenditure (CAPEX)
Installation Complexity	Ease of installation, commissioning, and operation
Reliability & Risk	System robustness and failure risk assessment

8. Deliverables

8.1. Primary Deliverable: PowerPoint Presentation Report

The main deliverable is a **professional-quality PowerPoint presentation report** that serves as the team's engineering submission. It must include:

1. System design overview and architecture
2. Complete Single Line Diagram (SLD) with annotations
3. Engineering calculations and analysis results
4. Simulation results with graphical outputs
5. Commercial comparison of starting methods
6. Final recommendation with supporting rationale

8.2. Supporting Deliverables

In addition to the main presentation, each group must submit:

- **Simulation files** – all MATLAB/Simulink or Python source files, clearly commented and organized
- **Supporting calculations** – handwritten or typed calculation sheets covering all required engineering analyses

9. Group Structure

Parameter	Requirement
Group Size	2 to 3 students (maximum)
Collaboration	All members must contribute to all sections
Submission	One submission per group

10. Grading Criteria

Project grades will be allocated according to the following weighted breakdown:

Grading Category	Weight (%)
Technical Design (system design, SLD, analysis, calculations)	**35%**
Simulation (modeling, results, and method comparison)	**15%**
Commercial Analysis (vendor selection, CAPEX, trade-off)	**15%**
Report & Files (presentation quality, clarity, completeness)	**35%**
Total	**100%**

11. Key Engineering Insight

Key Engineering Insight

The central challenge of this project is *not merely voltage conversion.* The two most critical engineering concerns are:

- **Managing high starting current** – which may reach 6× the rated current, causing severe voltage dips on an isolated diesel generator bus
- **Maintaining generator voltage stability** – ensuring the generator excitation and AVR can ride through the transient without tripping

A proven industry solution frequently employed in such scenarios combines a **step-up transformer** with an **MV Variable Frequency Drive (VFD)**, providing both voltage conversion and fully controlled soft starting.

12. Conclusion

Each group must deliver a clear, well-justified final recommendation that answers the following core project question:

Final Recommendation Question

What is the most practical, safe, and cost-effective solution to test a ***4 kV, 1.551 MW induction motor*** *using a* ***400 V diesel generator****, considering technical feasibility, generator stability, equipment cost, and installation complexity?*

The recommendation must be substantiated by the engineering analysis, simulation results, and commercial evaluation presented throughout the report, and must clearly identify the preferred starting method with specific vendor equipment proposed.

◇

Department of Electrical Engineering — EE360 Electric Energy Engineering
Term 252